

Using spherical coordinates:

- Jacobian of the transformation: $J = R^2 \sin \theta$
- Gradient:

$$\nabla(\vec{u}) = \begin{pmatrix} \frac{\partial u^r}{\partial r} & \frac{1}{R} \frac{\partial u^r}{\partial \theta} - \frac{u^\theta}{R} & \frac{1}{R \sin \theta} \frac{\partial u^r}{\partial \varphi} - \frac{u^\varphi}{R} \\ \frac{\partial u^\theta}{\partial r} & \frac{1}{R} \frac{\partial u^\theta}{\partial \theta} - \frac{u^r}{R} & \frac{1}{R \sin \theta} \frac{\partial u^\theta}{\partial \varphi} - \cot \theta \frac{u^\varphi}{R} \\ \frac{\partial u^\varphi}{\partial r} & \frac{1}{R} \frac{\partial u^\varphi}{\partial \theta} & \frac{1}{R \sin \theta} \frac{\partial u^\varphi}{\partial \varphi} + \cot \theta \frac{u^\theta}{R} + \frac{u^r}{R} \end{pmatrix}$$

- Kinematic assumption: $\vec{u} = u^\theta(\theta) \hat{e}_\theta = u(\theta) \hat{e}_\theta$
- Gradient for $\vec{u} = u(\theta) \hat{e}_\theta$: $\uparrow$ dropping θ in u

$$\nabla(\vec{u}) = \frac{1}{R} \begin{pmatrix} 0 & -u & 0 \\ 0 & \partial u / \partial \theta & 0 \\ 0 & 0 & \cot \theta u \end{pmatrix}$$

- Sym. gradient for $\vec{u} = u(\theta) \hat{e}_\theta$:

$$\varepsilon(\vec{u}) = \frac{1}{R} \begin{pmatrix} 0 & -u/2 & 0 \\ -u/2 & \partial u / \partial \theta & 0 \\ 0 & 0 & \cot \theta u \end{pmatrix}$$

- Geometrically nonlinear strain tensor:

$$\varepsilon^P(\vec{u}) = \varepsilon(\vec{u}) + \varepsilon^N(\vec{u})$$

$$\text{... and } \varepsilon^N(\vec{u}) = \frac{1}{2} (\nabla(\vec{u}) + \nabla(\vec{u})^T)$$

with $\varepsilon^N(\vec{u}) = \frac{1}{2}(\nabla(\vec{u}))^t \nabla(\vec{u})$

$$\varepsilon^N(\vec{u}) = \frac{1}{2R^2} \begin{pmatrix} 0 & 0 & 0 \\ -u & \partial u / \partial \theta & 0 \\ 0 & 0 & \cot \theta u \end{pmatrix} \begin{pmatrix} 0 & -u & 0 \\ 0 & \partial u / \partial \theta & 0 \\ 0 & 0 & \cot \theta u \end{pmatrix}$$

$$= \frac{1}{2R^2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & u^2 + (\partial u / \partial \theta)^2 & 0 \\ 0 & 0 & (\cot \theta)^2 u^2 \end{pmatrix}$$

- Weak form in spherical coordinates:

$$\int_S \text{div} \left(S \text{Id} + \lambda \text{tr}(\varepsilon^P(\vec{u})) \text{Id} + M \varepsilon^P(\vec{u}) \right) \cdot \vec{x} \, \text{Id} \, d\theta d\varphi =$$

$$\int_S \text{div} \left(\lambda \text{tr}(\varepsilon^P(\vec{u})) \text{Id} + M \varepsilon^P(\vec{u}) \right) \cdot \vec{x} \, \text{Id} \, d\theta d\varphi =$$

- Integration by parts, NO boundary term as S closed -

$$\int_S \left(\lambda \text{tr}(\varepsilon^P(\vec{u})) \text{Id} + M \varepsilon^P(\vec{u}) \right) : \varepsilon(\vec{x}) \, \text{Id} \, d\theta d\varphi =$$

$$2\pi \int_0^\pi \left(\lambda \text{tr}(\varepsilon^P(\vec{u})) \text{Id} + M \varepsilon^P(\vec{u}) \right) : \varepsilon(\vec{x}) \, \text{Id} \, d\theta \quad \forall \vec{x}$$

- The 2π cancels out in the weak problem
(both in left/right hand sides of equation).

- We need to compute:

- We need to compute :

$$\int_0^\pi \left(\Lambda \operatorname{tr}(\varepsilon^p(\vec{u})) \operatorname{Id} : \varepsilon(\vec{r}) \right) R^2 \sin \theta + \left(M \varepsilon^p(\vec{u}) : \varepsilon(\vec{r}) \right) R^2 \sin \theta \, d\theta$$

$$\underbrace{\left(\Lambda \operatorname{tr}(\varepsilon^p(\vec{u})) \operatorname{Id} : \varepsilon(\vec{r}) \right) R^2 \sin \theta :}$$

$$\operatorname{tr}(\varepsilon^p(\vec{u})) = \operatorname{tr}(\varepsilon(\vec{u})) + \operatorname{tr}(\varepsilon^N(\vec{u}))$$

$$\int \operatorname{tr}(\varepsilon(\vec{u})) = \frac{1}{R} \left(\frac{\partial u}{\partial \theta} + \cot \theta u \right)$$

$$\left[\operatorname{tr}(\varepsilon^N(\vec{u})) = \frac{1}{2R^2} \left[(1 + (\cot \theta)^2) u^2 + \left(\frac{\partial u}{\partial \theta} \right)^2 \right] \right]$$

$$\left(\operatorname{tr}(\varepsilon(\vec{u})) \operatorname{Id} : \varepsilon(\vec{r}) \right) R^2 \sin \theta =$$

$$\left\{ \frac{\partial u}{\partial \theta} \frac{\partial v}{\partial \theta} + \cot \theta u \frac{\partial v}{\partial \theta} + \frac{\partial u}{\partial \theta} \cot \theta v + (\cot \theta)^2 u v \right\} \sin \theta$$

$$\left(\operatorname{tr}(\varepsilon^N(\vec{u})) \operatorname{Id} : \varepsilon(\vec{r}) \right) R^2 \sin \theta =$$

$$\frac{1}{2R} \left\{ \left[(1 + (\cot \theta)^2) u^2 + \left(\frac{\partial u}{\partial \theta} \right)^2 \right] \left[\frac{\partial v}{\partial \theta} + \cot \theta v \right] \right\} \sin \theta$$

$$\left(\Lambda \operatorname{tr}(\varepsilon^p(\vec{u})) \operatorname{Id} : \varepsilon(\vec{r}) \right) R^2 \sin \theta =$$

$$\Lambda \left\{ \frac{\partial u}{\partial \theta} \frac{\partial v}{\partial \theta} + \cot \theta u \frac{\partial v}{\partial \theta} + \frac{\partial u}{\partial \theta} \cot \theta v + (\cot \theta)^2 u v \right\} \sin \theta +$$

$$\frac{\Lambda}{2R} \left\{ \left[(1 + (\cot \theta)^2) u^2 + \left(\frac{\partial u}{\partial \theta} \right)^2 \right] \left[\frac{\partial v}{\partial \theta} + \cot \theta v \right] \right\} \sin \theta$$

$$(M \mathcal{E}^P(\vec{u}) : \mathcal{E}(\vec{v})) R^2 \sin \theta :$$

$$\mathcal{E}(\vec{u}) : \mathcal{E}(\vec{v}) R^2 \sin \theta =$$

$$\left\{ \frac{uv}{2} + \frac{\partial u}{\partial \theta} \frac{\partial v}{\partial \theta} + (\cot \theta)^2 uv \right\} \sin \theta$$

$$\mathcal{E}^N(\vec{u}) : \mathcal{E}(\vec{v}) R^2 \sin \theta =$$

$$\frac{1}{2R} \left\{ \left[u^2 + \left(\frac{\partial u}{\partial \theta} \right)^2 \right] \frac{\partial v}{\partial \theta} + (\cot \theta)^3 u^2 v \right\} \sin \theta$$

$$(M \mathcal{E}^P(\vec{u}) : \mathcal{E}(\vec{v})) R^2 \sin \theta =$$

$$M \left\{ \frac{uv}{2} + \frac{\partial u}{\partial \theta} \frac{\partial v}{\partial \theta} + (\cot \theta)^2 uv \right\} \sin \theta +$$

$$\frac{M}{2R} \left\{ \left[u^2 + \left(\frac{\partial u}{\partial \theta} \right)^2 \right] \frac{\partial v}{\partial \theta} + (\cot \theta)^3 u^2 v \right\} \sin \theta$$

- Gateaux derivative of the weak form (nonlinear terms):

$$a_{NL}^{\Lambda}(u, v) = \frac{\Lambda}{2R} \left\{ \left[\underbrace{(1 + (\cot \theta)^2)}_{= (\csc \theta)^2} u^2 + \left(\frac{\partial u}{\partial \theta} \right)^2 \right] \left[\frac{\partial v}{\partial \theta} + \cot \theta v \right] \right\} \sin \theta$$

$$a_{NL}^{\Lambda}(u + \varepsilon du, v) = \underbrace{(\ast)}_{=}$$

$$\begin{aligned}
 \hat{a}_{NL}(u + \varepsilon du, \nu) &= \\
 \frac{\Lambda}{2R} \left\{ \left[\csc^2 \theta (u + \varepsilon du)^2 + \left(\frac{\partial u}{\partial \theta} + \varepsilon \frac{\partial du}{\partial \theta} \right)^2 \right] \overbrace{\left[\frac{\partial \nu}{\partial \theta} + \cot \theta \nu \right]}^{(*)} \right\} \sin \theta &= \\
 \frac{\Lambda}{2R} \left\{ \left[\csc^2 \theta (u^2 + 2u\varepsilon du + \varepsilon^2 du^2) + \left(\frac{\partial u^2}{\partial \theta} + 2\varepsilon \frac{\partial u}{\partial \theta} \frac{\partial du}{\partial \theta} + \varepsilon^2 \frac{\partial du^2}{\partial \theta} \right) \right] (*) \right\} \sin \theta
 \end{aligned}$$

$$\begin{aligned}
 \left(\hat{a}_{NL}(u + \varepsilon du, \nu) - \hat{a}_{NL}(u, \nu) \right) / \varepsilon &= \\
 \frac{\Lambda}{2R} \left\{ \left[\csc^2 \theta (2u du + \varepsilon du) + \left(2 \frac{\partial u}{\partial \theta} \frac{\partial du}{\partial \theta} + \varepsilon \left(\frac{\partial du}{\partial \theta} \right)^2 \right) \right] (*) \right\} \sin \theta
 \end{aligned}$$

$$\begin{aligned}
 \lim_{\varepsilon \rightarrow 0} \left(\hat{a}_{NL}(u + \varepsilon du, \nu) - \hat{a}_{NL}(u, \nu) \right) / \varepsilon &= \\
 \frac{\Lambda}{R} \left\{ \left[\csc^2 \theta u du + \frac{\partial u}{\partial \theta} \frac{\partial du}{\partial \theta} \right] \left[\frac{\partial \nu}{\partial \theta} + \cot \theta \nu \right] \right\} \sin \theta
 \end{aligned}$$